CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 5: Arithmetic Progressions

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the common difference and the next two terms of the AP $-5, -1, 3, 7, \ldots$
2. Find the 25th term of the AP $12, 7, 2, \ldots$
3. Which term of the AP $4, 9, 14, \ldots$ is 254?
4. The 5th term of an AP is 20 and the 12th term is 48. Find the first term and the common difference.
5. How many two-digit numbers are divisible by 4?
6. Find the 6th term from the end of the AP $8, 15, 22, \ldots, 148$.
7. Find the sum of the first 30 terms of the AP $2, 6, 10, \ldots$
8. Find the sum of all odd numbers between 0 and 60.
9. How many terms of the AP $20, 18, 16, \ldots$ must be added to get a sum of 110? Explain the two answers.
10. A sum of ₹1000 is to be paid in 10 instalments that form an AP. If the first instalment is ₹55, find the common difference and the last instalment.

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Answer Key

1. $d = 4$; next terms 11 and 15
2. $12 + 24 \times (-5) = -108$
3. The 51st term
4. $a = 4, d = 4$
5. $12, 16, \ldots, 96$: 22 numbers
6. $148 - 5 \times 7 = 113$
7. $15\,[\,4 + 116\,] = 1800$
8. $1 + 3 + \cdots + 59 = 30^2 = 900$
9. $n = 10$ or $n = 11$; the 11th term is 0, so both sums are 110.
10. $d = 10$; last instalment ₹145

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